

PALACIOS, TX, USA

WMO: 722555

Lat: 28.725N

Lon: 96.254W

Elev: 12

StdP: 14.69

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 12935

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	33.1	36.4	20.8	15.6	43.1	25.2	19.3	45.5	28.3	51.2	26.2	50.9	11.6	0	0.493

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	93.1	78.8	91.7	79.0	90.7	78.9	81.8	88.8	81.1	88.2	80.4	87.4	11.9	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.8	154.6	85.2	79.2	151.6	84.9	78.7	148.8	84.6	45.3	89.3	44.6	88.0	44.0	87.5	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.5	21.0	19.1	27.3	97.6	3.6	3.2	24.8	99.9	22.6	101.8	20.6	103.6	18.0	106.0		
			25.2	82.8	3.4	1.1	22.7	83.6	20.8	84.3	18.9	84.9	16.4	85.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	71.2	55.3	58.8	64.3	70.7	77.7	82.9	84.6	84.7	80.7	73.2	64.3	57.1
	DBStd	12.46	9.03	9.03	7.81	5.99	4.47	2.94	2.32	2.79	4.42	7.02	8.61	9.17
	HDD50	128	51	28	7	0	0	0	0	0	0	0	7	35
	HDD65	1068	319	205	109	22	1	0	0	0	0	18	116	278
	CDD50	7882	214	275	450	621	857	987	1071	1076	920	720	437	254
	CDD65	3349	17	32	88	193	394	537	606	611	470	273	96	32
	CDH74	34194	23	48	216	952	3460	6328	7642	7630	4929	2385	499	82
	CDH80	13333	0	2	9	96	832	2576	3478	3633	2002	656	46	3
Wind	WSAvg	10.5	10.6	11.2	11.7	12.3	12.0	10.7	9.9	9.5	8.6	9.5	9.7	10.1
Precipitation	PrecAvg	46.9	3.3	2.4	2.6	2.9	4.7	5.0	3.8	3.7	7.0	4.2	3.3	3.0
	PrecMax	66.3	10.3	6.9	9.3	9.3	13.2	15.6	12.2	9.2	23.7	13.3	9.2	8.2
	PrecMin	27.7	0.3	0.2	0.2	0.1	0.0	0.0	0.1	0.1	0.6	0.6	0.1	0.7
	PrecStd	11.2	2.2	1.7	2.2	2.3	3.7	4.1	3.2	2.7	5.8	3.1	2.5	2.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.9	78.2	81.1	85.6	89.1	93.8	93.8	95.7	95.0	90.3	83.6	79.4
		MCWB	66.7	68.9	69.5	68.9	76.8	78.0	78.1	78.6	77.7	76.6	74.8	71.5
	2%	DB	73.5	75.2	78.5	82.3	87.2	91.1	91.9	93.3	91.9	87.7	81.3	76.3
		MCWB	67.3	68.6	69.8	72.0	76.5	78.4	79.3	79.4	78.0	76.5	73.2	70.1
	5%	DB	71.6	73.2	76.5	80.7	85.8	90.0	90.9	92.0	90.4	85.8	79.2	73.6
		MCWB	66.5	67.5	69.4	72.2	76.1	78.6	79.4	79.4	77.8	75.5	71.4	68.7
	10%	DB	69.1	71.1	74.5	79.0	84.3	88.7	90.1	90.8	88.7	83.8	77.0	71.4
		MCWB	65.0	66.7	68.7	71.6	75.7	78.4	79.3	79.3	77.4	74.7	70.6	67.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.2	72.9	73.5	76.9	80.4	82.8	82.3	82.4	82.3	81.0	77.7	73.7
		MCDB	74.0	75.6	76.7	80.5	85.0	87.7	89.6	90.8	88.6	85.4	80.9	76.5
	2%	WB	69.5	71.1	72.5	75.6	79.2	81.3	81.4	81.6	81.0	79.6	75.6	72.0
		MCDB	71.8	73.5	75.5	79.4	83.7	87.2	89.1	89.7	87.1	83.9	78.8	74.7
	5%	WB	68.1	69.5	71.4	74.6	78.2	80.4	80.6	80.9	80.2	78.2	74.0	70.4
		MCDB	70.5	71.7	74.3	78.3	83.1	86.5	88.1	88.8	86.5	82.9	77.4	72.7
	10%	WB	66.1	68.1	70.4	73.5	77.4	79.9	80.2	80.3	79.4	77.1	72.4	68.6
		MCDB	68.3	70.1	73.4	77.1	82.5	86.0	87.5	88.0	85.8	81.9	75.8	70.8
Mean Daily Temperature Range	5% DB	MDBR	17.6	16.3	15.6	14.7	12.4	11.6	11.2	12.8	15.4	17.7	17.9	17.8
		MCDBR	17.0	15.4	13.9	14.2	12.6	12.6	12.2	13.9	16.3	16.4	16.5	17.5
		MCWBR	12.4	10.1	8.1	7.7	5.5	4.8	4.4	4.8	6.3	7.7	9.9	12.5
	5% WB	MCDBR	14.9	12.9	11.7	11.0	10.2	10.4	10.9	12.2	13.0	13.1	14.0	15.6
		MCWBR	12.5	10.0	7.9	6.9	5.3	4.9	4.5	4.9	6.0	7.8	10.4	12.6
Clear-Sky Solar Irradiance	taub	0.351	0.364	0.382	0.441	0.491	0.460	0.485	0.464	0.451	0.391	0.381	0.349	
	taud	2.497	2.469	2.409	2.244	2.136	2.288	2.232	2.293	2.296	2.457	2.428	2.517	
	Ebn at Noon	282	287	288	272	258	265	258	263	263	274	268	277	
	Edh at Noon	29	32	36	44	50	43	45	42	41	33	31	28	
All-Sky Solar Radiation	RadAvg	929	1077	1446	1724	1948	2102	2054	1956	1641	1426	1073	840	
	RadStd	99	174	130	150	125	131	186	113	150	148	117	102	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.61	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+247	+194
(47)	Regional (34 neighbors)	+0.59	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+211	+180

Nomenclature: See separate page